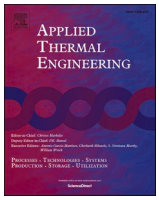
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Research Paper



Numerical study on electrical and thermal performance of a novel concentrated Photovoltaic/Thermal system

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ABSTRACT

Increasing thermal output while maintaining high electrical efficiency of photovoltaic modules is one of the current research focuses for concentrated photovoltaic/thermal collectors. In this study, a concentrated photovoltaic/thermal system partially filled with porous media in the pipes and cooled by micro-encapsulated phase change material slurry is numerically investigated. Micro-encapsulated phase change material slurry and porous media are used for the first time in a concentrated photovoltaic/thermal solar collector to increase the overall efficiency of the system. In addition, a gradient porous media with optimized design and arrangement is used to maximize the overall system efficiency. The proposed system is numerically investigated by utilizing the discrete ordinates radiation model and the energy equation at local thermal equilibrium and rigorously validated against published results. The performance of the concentrated photovoltaic/thermal system is investigated by varying the concentration ratio, the melting point, latent heat and volume fraction of the micro-encapsulated phase change material slurry, as well as the filling method, thickness and porosity of the porous media. The results show that the performance of the concentrated photovoltaic/thermal system can be improved apparently by using micro-encapsulated phase change material slurry and partially filled with porous medium. The thermal and electrical efficiencies of the system can be increased by 5.98 % and 1.26 %, respectively, by employing micro-encapsulated phase change material slurry instead of water as heat transfer fluid (with volume fraction of 0.15 and inlet velocity of 0.8 m/s). The thermal and electrical efficiencies of the system can be increased by 2.42 % and 8.82 %, respectively, by filling with a porous media with layer thickness of 4.76 mm (with porosity of 0.7 and pore size of 0.001 m, and inlet velocity of 0.3 m/s). However, partially filled with porous medium along the pipe axis has a negative impact on the system performance. This study provides a route for concentrated photovoltaic/thermal collectors that employ smart, multi-functional fluids offering superior electrical and thermal performances.

1. Introduction

Energy consumption has increased internationally over the last two decades, owing to population expansion and improvement of living conditions. The widespread use of fossil energy has resulted in ever-increasing pollution and global warming issues, which substantially impede sustainable and healthy economic development over the world. Researchers have been relentlessly in their pursuit of renewable energy in order to strengthen the energy supply and alleviate the pressures of rising energy consumption. Solar energy, as one of the most promising renewable energy, has received great attention in recent decades due to its vast reserves and environmental friendliness [1–3]. One of the most common methods of harnessing solar energy is photovoltaic power generation. However, only around 20 % of solar radiation landing on PV cells can currently be converted into electricity, with the remainder being wasted as heat. Researchers have suggested to incorporate optical elements (e.g., Fresnel lens [4], composite parabolic concentrators [5], V-trough concentrators [6], et al.) into PV/T collectors to construct CPV/T to boost solar energy utilization and system efficiency. Compared to a flat panel PV/T, the CPV/T system can significantly increase the amount of incident solar radiation, lowering costs while increasing thermal output [7,8]. Nevertheless, an increase in global temperatures can be detrimental to the durability and performance of PV modules. One of the current research focuses is to limit the CPV module temperature to a reasonable working temperature range [9–12].

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Nomenclature

A	Area, m^2
CPC	Compound parabolic concentrator
CPV/T	Concentrated photovoltaic/thermal
CR	Concentration ratio
D	Diameter, m
G	Solar irradiation, W/m^2
F	Packing factor
PDSD	Porosity decreasing and pore size decreasing
PDSI	Porosity decreasing and pore size increasing
PISD	Porosity increasing and pore size decreasing
PISI	Porosity increasing and pore size increasing

Subscripts

elec	Electrical
f	Fluid
pa	Particle
po	Pore
s	MPCM slurry
th	Thermal
w	Water

Greek

α	Absorptivity
β	Temperature coefficient, K^{-1}
ε	Emissivity
k	Thermal conductivity
n	Refractive index
τ	Transmissivity
ω	Porosity of porous media
γ	Reflectivity
φ_V	Volume fraction
φ_m	Mass fraction

In comparison to traditional cooling fluids, phase change material (PCM) may absorb or release enormous amounts of heat at nearly constant temperature. This technology has been extensively used for cooling electronic components, buildings, and others [13–15]. PV modules may store thermal energy while cooling by combining PCM and CPV/T. Mohamed et al. [16] designed innovative PCM-integrated CPV modules. The simulation results revealed that the proposed triple package design reduced the average temperature of the PV cell by 16 °C while increasing the system's electrical and thermal efficiency by 21.3 % and 5.29 %, respectively, compared to the traditional CPV system. Nasef et al. [17] used a combination of PCM and nanofluid to cool the CPV/T collector. According to the modelling results, using PCM and nanofluid enhanced the output of the CPV assembly by 2.7 %. However, the inherent flaws of conventional solid–liquid PCM, such as its propensity to escape and incompatibility with surrounding materials, limit its future applications. Researchers have therefore proposed microencapsulating PCM in order to create a microencapsulated phase-change suspension. As a high-performance fluid for heat transfer, the evident thermal conductivity and effective specific heat capacity of MPCM slurry are significantly increased [18]. MPCM slurry has been used to cool flat-plate PV/T collectors [19–21]. Yu et al. [22] investigated the influence of MPCM content and melting temperature on PV/T system efficiency. They concluded that the increase in concentration is advantageous for the system's energy and exergy efficiency. While altering the melting temperature necessitates a trade-off between energy and exergy efficiency. Liu et al. [23] compared an MPCM slurry-cooled CPV/T collector to a standard water-cooled CPV/T system. The simulation results highlighted the impact of MPCM slurry on the overall system performance.

Porous media are excellent materials for enhancing heat transfer efficacy. In the majority of current research involving the cooling of PV modules, porous media such as metal foams are combined with PCM to compensate for PCM's poor thermal conductivity [24,25]. Rahmanian et al. [26] compared the effects of nanoparticles and metal foams on the enhancement of PCM. Results demonstrated that nano-PCM sink substantially contributes to the cooling effect of metal foam-PCM sink on CPV components. Duan et al. [27] studied the variation of electrical characteristics of a CPV-PCM system with variable porosity and height of metal foam. The simulation results demonstrated that the electrical efficiency of the system increases as the porosity decreases, while there is an optimal value for the height of the PCM-porous to achieve the greatest electrical efficiency of the CPV. Using porous media as fins inside the tube to investigate the cooling effect of PV/T systems is another common research direction [28,29]. Fatih et al. [30] compared the effectiveness of three cooling methods, including conventional fins, hollow fins, and wavy tubes with porous layers. The simulation results revealed that channel cooling with a porous layer and an undulating wall provided the most effective cooling. Specifically, when nanofluid is used as the cooling fluid, the PV plate can achieve a temperature decrease of 37.6 K. Mohamed et al. [31] have proposed an innovative system based on a porous deflector PV/T-nanofluid. The optimal cooling performance is obtained when five porous deflectors with an aspect ratio of one are installed in the channel at a nanofluid volume fraction of 3 %. In addition, based on the excellent performance of porous media and the popularity of additive manufacturing technology, gradient porous media with unique properties are gradually used in many fields. Shi et al. [32] compared four configurations in which the pore size and porosity varied along the radial direction in the case of partially filled pipes with porous media, respectively. The results showed that the flow resistance of the gradient porous media can be reduced by up to 19.573 % and the heat transfer efficiency can be improved by up to 7.088 % compared to the homogeneous porous media when the filling ratio is 0.7.

By the literature survey, it can be seen that MPCM slurry and porous media can play an effective role in improving the heat transfer capability and overall performance of CPV/T systems when working as heat storage media and heat carriers. Gradient porous media can also exhibit better performance than uniform porous media. However, the use of porous media, especially gradient porous media, and MPCM slurry as a combination of heat transfer fluids in CPV/T systems is currently less addressed in the existing literature. The placement of porous media with high thermal conductivity in the pipeline increases the contact area between the fluid and the channel matrix and causes perturbation to the fluid near the near-wall surface, thus mitigating the adverse effects due to the increase in the viscosity of the MPCM slurry. Therefore, this work evaluates the feasibility and effectiveness of improving the overall performance of the CPV/T system by completely or partially filling the pipeline with a porous medium and using MPCM slurry as a coolant. The forthcoming results may bring new perspectives for heat transfer enhancement technologies. For a clearer comparison, Table 1 summarizes the latest researches on heat transfer enhancement of systems using MPCM slurry or porous media in CPV/T systems.

A three-dimensional model of a CPV/T system is constructed based on a composite paraboloidal concentrator (CPC). This work examines the effects of the concentration ratio, coolant flow rate, and MPCM slurry characteristics (i.e., melting point, latent heat, and volume fraction) in the presence of a smooth channel and compares them to those of conventional CPV/T collectors. After identifying the optimal parameters, porous layers are designed and installed in the pipeline to improve heat conduction. In this paper, two kinds of partial filling of a porous layer are examined: gradual thickness increases along the pipe wall and

Table 1
Summary of the latest studies on CPV/T-MPCM or CPV/T-porous media system.

Ref.	Means of enhancing heat transfer	Whether to consider radiation	Performance enhancement	Major conclusions
Eisapour et al. [20]	MPCM slurry	No	$\Delta\,\eta_{elec} = 0.4$ %	The effect played by nanoparticles in MPCM slurry depends specifically on the thermal conductivity and heat capacity of the nanoparticles.
Yu et al. [22]	MPCM slurry	No	$\Delta\,\eta_{o} = 8.3$ % $\Delta\eta_{o,\ exergy} = 3.23$ %	A lower melting temperature resulted in higher energy efficiency, whereas a higher melting temperature is helpful for exergy efficiency improvement.
Liu et al. [23]	MPCM slurry	No	$\Delta\,\eta_{o} = 1.8$ %	MPCM thermal conductivity does not have significant effect on PV/T performance.
Hamada et al. [33]	MPCM slurry	No	$\Delta T_{PV} = 10$ ℃ $\Delta\eta_{th,\ exergy} = 208$ %	MPCM slurry effectively filters out incoming solar radiation.
Farideh et al. [34]	MPCM slurry	No	$\Delta T_{PV} = 33.284$ ℃	Nanofluid-MPCM filter is not suitable for concentration ratios of less than 10.
Pawar et al. [35]	Homogeneous porous media	No	$\Delta\,\eta_{o} = 10.62$ %	Aluminum foam shows more heat transfer enhancement than copper foam.
Asefi et al. [25]	Homogeneous porous media	No	$\Delta\,\eta_{th} = 43.1$ % $\Delta\,\eta_{elec} = 1.5$ % $\Delta\eta_{o,\ exergy} = 1.16$ %	The combination of copper foam and Ag/water nanofluid shows the highest thermal, electrical and exergy efficiencies.

gradual thickness increases along the pipe axis. Additionally, the effects of the porosity and pore size of the porous layer and the coolant flow rate on the performance of the CPV/T system are evaluated. Finally, the porosity and pore size of the porous media are varied linearly along the conduit diameter to discover the solution that maximizes the overall efficiency of the system. The study can provide valuable guidance for the optimal configuration of CPV/T collectors and the comprehension of the performance enhancement mechanism.

2. Description of the CPV/T system

Fig. 1 gives the schematic diagram of the CPV/T model used in this work. The composite paraboloidal concentrator (CPC) is widely used in photovoltaics because it has a larger acceptance angle and thus receives a wider spectrum of radiation than other concentrators [36]. In this version, the CPC is mounted on the upper portion of the PV module in order to direct incident solar radiation towards the receiver. In addition, from top to bottom, there are a glass layer, air gap, EVA-a, PV cell, EVA-b, absorber plane, and ten straight pipelines that are distributed uniformly below. The EVA layer protects the solar cell against UV radiation and moisture [37]. The pipes are welded to the back of the absorber plate using copper bonding. MPCM slurry flowing through the pipelines, and absorbs the surplus of heat produced by the PV module, which is then utilized for thermal purposes. Due to the symmetry of the model, only one-half of the system is considered as the computational domain for modelling.

Fig. 2 displays the specific model for the ten pipes containing porous media. The cases M0 and M5 represent the smooth pipe and the fully filled pipe by porous media, respectively. The ratio of the thickness of the porous layer to the inner diameter of the conduit is $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, and $\frac{3}{4}$ in descending order for cases M1 through M4. M6-M9 represent the cases of laying porous media along the pipe axis, and the thickness of the porous layer in relation to the interior diameter of the pipe is $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, and $\frac{3}{4}$, respectively.

The dimensions, thermophysical properties, and optical properties of

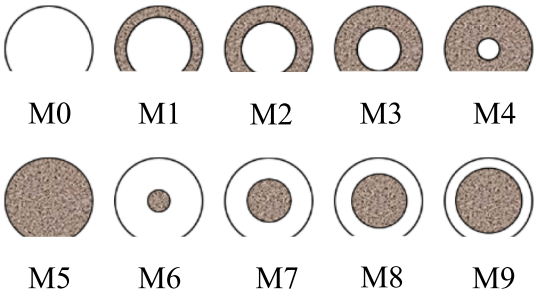


Fig. 2. Schematic diagram of porous pipe structure.

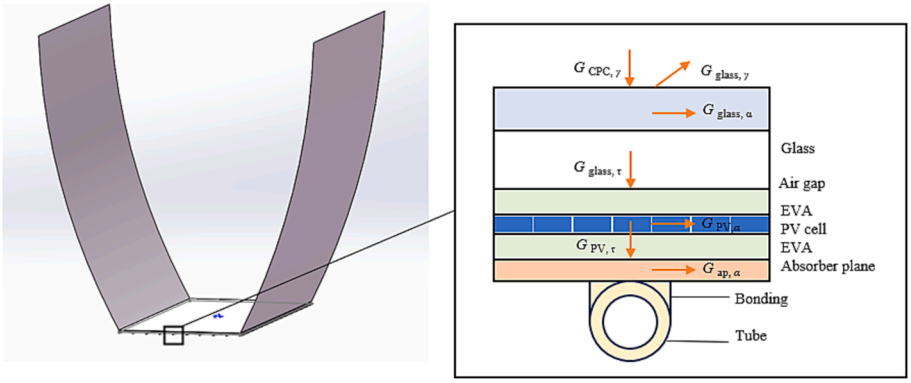


Fig. 1. Schematic diagram of CPV/T system.

Table 2
The properties of the materials used in the simulation.

Components	Dimensions (mm) (length × width × height)	Thermophysical properties	Optical properties	Others	Ref.
CPC (Aluminum)	2000 × 2400 × 2270.2	$\rho = 2719$ kg/m^3 $C_p = 871$ J/(kg·K) $k = 401$ W/(m·K)	$\alpha = 0.2$ $\tau = 0$		[38]
Tube (Copper)	$D_{out} = 11.32$ mm, $D_{in} = 9.52$ mm	$\rho = 8978$ kg/m^3 $C_p = 381$ J/(kg·K) $k = 387.6$ W/(m·K)	–		[38]
Glass pattern	2000 × 1000 × 4	$\rho = 2650$ kg/m^3 $C_p = 786$ J/(kg·K) $k = 1.5$ W/(m·K)	$\alpha = 0.08$ $\tau = 0.905$ $\varepsilon = 0.88$ n = 1.526		[39]
Air gap	2000 × 1000 × 12	$\rho = 1.225$ kg/m^3 $C_p = 1006.43$ J/(kg·K) $k = 0.0242$ W/(m·K)	$\tau = 1$		[39]
EVA	2000 × 1000 × 0.5	$\rho = 960$ kg/m^3 $C_p = 2090$ J/(kg·K) $k = 0.35$ W/(m·K)	$\tau = 1$		[39]
PV cell (Silicon)	2000 × 1000 × 0.3	$\rho = 2330$ kg/m^3 $C_p = 700$ J/(kg·K) $k = 100$ W/(m·K)	$\alpha = 0.93$ $\tau = 0.07$ $\varepsilon = 0.9$ n = 3.69	$T_{ref} = 298.15$ K $\eta_{ref} = 0.15$ $\beta_{ref} = 0.0045$ K^{-1} $F = 0.9$	[39]
Absorber plane (Copper)	2000 × 1000 × 0.2	$\rho = 8978$ kg/m^3 $C_p = 381$ J/(kg·K) $k = 387.6$ W/(m·K)	$\alpha = 0.95$ $\tau = 0.05$		[39]
MPCM	$D_{pa} = 10$ μm	$\rho = 946.4$ kg/m^3 $C_p = 1973.1$ J/(kg·K) $k = 0.749$ W/(m·K) $T_m = 310.1 \sim 311.3$ K $\Delta H = 192.66$ (J/kg)	–		[20]
Porous layer (Copper)	$D_{po} = 1$ mm	$\rho = 8978$ kg/m^3 $C_p = 381$ J/(kg·K) $k = 387.6$ W/(m·K)	–	–	[38]

the different components are listed in Table 2. The path of the solar radiation after reflection and absorption by the concentrator, after reaching the glass cover, within the system components is shown in Fig. 1. Most of the solar irradiance that passes through the glass layer is transmitted, and a small portion is reflected or absorbed, depending on the optical properties of the glass. In this study, both the air gap and EVA layers are considered to be 100 % transmission, i.e., no reflection or absorption. The transmitted irradiance reaching the PV module is absorbed and part of the absorbed radiation is converted into electrical energy in the PV cell and the rest is converted into thermal energy.

3. Mathematical models

3.1. Assumptions and governing equations

The following assumptions are made in construction of the mathematical model in this study.

(1) The bottom of the tube and the side walls of the module are considered adiabatic.
(2) The incident solar radiation is perpendicular to the surface of the PV module [40].
(3) A perfect thermal contact is considered between each solid layer, i.e., the thickness of the bonding material between the layers is ignored.
(4) Water and phase change fluids used in the model are treated as fully developed, incompressible Newtonian fluids.
(5) MPCM slurry flows in the tube with a particle distribution that can be regarded as uniform, and the effect of particle settling on the thermophysical properties of the suspension is ignored.
(6) The porous media within the conduit are isotropic and in local thermal equilibrium with the fluid.

On the basis of the aforementioned hypotheses, the mass, momentum, and energy equations used in this investigation are defined as follows,

Continuity:

$$\nabla\cdot\left(\rho_f \vec{V}_f\right) = 0 \tag{1}$$

Momentum:

$$\nabla\cdot\left(\rho_f \vec{V}_f \vec{V}_f\right) = -\nabla\vec{P} + \mu_f\left(\nabla^2\vec{V}_f\right) - \rho_f\vec{g} \tag{2}$$

Energy:

$$\nabla\cdot\left(\left((1-\omega)(\rho C_p)_f + \omega(\rho C_p)_{porous}\right)\vec{V}_f T\right) = k_{eff}\nabla^2 T \tag{3}$$

in Eqs. (1), (2), and (3), ρ, $\vec{V}$, $\vec{P}$, and T are density, velocity, pressure and temperature, respectively. Subscript f refers to fluid (water or MPCM slurry). μ, ω, and C_p are viscosity, porosity of porous media and specific heat capacity, respectively. k_{eff} is effective thermal conductivity, which is calculated from Eq. (4).

$$k_{eff} = (1-\omega)k_f + \omega k_{porous} \tag{4}$$

The MPCM slurry used in this investigation is composed of n-eicosane microcapsules and water. The diameter of MPCM particle is set to be 10 μm, while the phase change temperature range is set to be 310.1–311.3 K in the present study. Other important property parameters are detailed in Table 2. Based on assumption (6), the following equation is used to calculate the properties of MPCM slurry:

According to the mass balance, the density of MPCM slurry can be expressed as

$$\rho_s = (1-\varphi_V)\rho_w + \varphi_V\rho_{mpcm} \tag{5}$$

where φ_V refers to volume fraction of MPCM slurry. Subscript s and w are MPCM slurry and water, respectively.

The static thermal conductivity of MPCM slurry can be derived from the Maxwell model [41].

$$k_s = k_w \frac{2 + \frac{k_{mpcm}}{k_w} + 2\varphi_V\left(\frac{k_{mpcm}}{k_w} - 1\right)}{2 + \frac{k_{mpcm}}{k_w} - \varphi_V\left(\frac{k_{mpcm}}{k_w} - 1\right)} \tag{6}$$

The static thermal conductivity formula assumes that there is no slip between the base fluid and the particles. However, the relative motion that exists between the particle–particle, particle-base fluid, and particl-wall surfaces all result in an increase in the overall thermal conductivity. It is common to express the effective thermal conductivity of MPCM slurry by the Eq. (7) [42].

$$k_f = k_{s,eff} = k_s\left(1 + 3\varphi_V \mathrm{Pe}_p^{1.5}\right) \tag{7}$$

The specific heat capacity of MPCM slurry is calculated by Eq. (8) [43].

$$C_p(T) = \begin{cases} C_{p,\ \mathrm{MPCM}} & T < T_m - \Delta T \\ C_{p,\ \mathrm{MPCM}} + \dfrac{L}{\Delta T} & T_m - \Delta T \le T < T_m + \Delta T \\ C_{p,\ \mathrm{MPCM}} & T \ge T_m + \Delta T \end{cases} \tag{8}$$

where L refers to latent heat of MPCM particle.

The viscosity of MPCM slurry is usually described by the following correlation equation, which has been validated for MPCM slurry with particle sizes in the range of 3–300 μm and volume fractions of less than 20 % [44]. It considers the MPCM slurry as a homogeneous fluid.

$$\frac{\mu_s}{\mu_f} = \left(1 - \varphi_V - 1.16{\varphi_V}^2\right)^{-2.5} \tag{9}$$

Two important parameters of porous materials, the viscous drag coefficient C_1 and the inertial drag coefficient C_2, are calculated from Eq. (10) and (11), respectively [45]. These two semi-empirical correlations hold on the assumption that the porous media is homogeneous and filled with spherical particles.

$$C_1 = \frac{150(1-\omega)^2}{D_{po}^2\omega^3} \tag{10}$$

$$C_2 = \frac{3.5(1-\omega)}{D_{po}\omega^3} \tag{11}$$

where D_{po} refers to pore size of porous media.

Finally, in order to describe the overall performance of the CPV/T system, the thermal, electrical and overall efficiencies of the system are defined as [46,47],

$$\eta_{th} = \frac{\dot{m}\cdot\int_{T_{in}}^{T_{out}} C_{p,s}(T)dT}{G_{sun}A_rCR} \tag{12}$$

$$\eta_{elec} = \frac{E_{elec} - \frac{\dot{m}\Delta P}{\rho_s\eta_{Pump}}}{G_{sun}A_rCR} \tag{13}$$

$$\eta_o = \eta_{th} + \eta_{elec} \tag{14}$$

In Eqs. (12) and (13), $\dot{m}$, T_{out}, and T_{in} are mass flow rate, inlet temperature and outlet temperature of heat transfer fluid. G_{sun}, A_r, and η_{Pump} are solar incident radiation, area of receiver, and pump efficiency (η_{Pump} = 0.8). CR refers to concentration ratio of CPC, which is calculated from Eq. (15). E_{elec} refers to power output of solar cell, which is calculated from Eq. (16).

$$CR = \frac{A_a}{A_r} \tag{15}$$

$$E_{elec} = G_{sun}\cdot\eta_{ref}(1 - \beta_{ref}\cdot(T_{PV} - T_{ref})\cdot F) \tag{16}$$

where A_a refers to aperture area. η_{ref}, β_{ref}, T_{ref}, and F are reference cell efficiency, reference temperature coefficient, reference temperature, and packing factor.

3.2. Simulated method and mesh independence test

The above mass, momentum, and energy equations are solved by the computational fluid dynamics software FLUENT for effective multi-physics field simulation. In the present study, the radiative transfer term is solved using the DO radiation model. When the fluid flow state in the tube is turbulent, the k-ε turbulence model is used for calculation. The equations about the *k-ε* turbulence model are shown in Eqs. (17) and (18). Since the y + value is close to 1, the Enhances Wall function is chosen.

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\left(\mu + \frac{\mu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right) + G_k + G_b - \rho\varepsilon - Y_M \tag{17}$$

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \frac{\partial(\rho\varepsilon u_i)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(\left(\mu + \frac{u_t}{\sigma_\varepsilon}\right)\frac{\partial\varepsilon}{\partial x_j}\right) + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon}G_b) - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} - R_\varepsilon \tag{18}$$

where, G_k represents the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. $C_{1\varepsilon}$, $C_{2\varepsilon}$, and $C_{3\varepsilon}$ are constants, the default values are 1.44, 1.92 and 0.09, respectively. σ_k and σ_ε are the turbulent Prandtl numbers for k and ε, respectively.

By constructing the Mixture model and ignoring the slip velocity between the particles and the base fluid, the flow of the MPCM suspension inside the cooling tube is simulated. The interior of the CPC is defined as ideal air (α = 0), concentrating all solar radiation on the PV module. The primary causes of heat loss from the upper glass cover are convection and radiation, and the convective heat transfer coefficient and sky temperature are defined as [48,49], respectively.

$$h_{conv} = 5.7 + 3.8U_{wind} \tag{19}$$

$$T_{sky} = 0.0522T_{amb}^{1.5} \tag{20}$$

where U_{wind} and T_{amb} are wind speed and ambiance temperature.

A pressure-based solver and pressure and velocity coupled scheme are used in the simulation. The pressure correction scheme is in "PRESTO!" format. The "Second Order Upwind" scheme is chosen to discretize the convective, diffusive, and radiative terms. The residuals of convergence are set as 1e-4 for the continuity and momentum equations and 1e-7 for the energy and DO equations.

The CPV/T system with a flow rate of 0.4 m/s for MPCM slurry is firstly examined in order to get the reasonable independent mesh. Five different sets of meshes (2223084, 2426526, 2824400, 3289452, 3697720) are subjected to mesh independence tests as given in Fig. 3. It is shown that 2,824,400 is acceptable mesh sizes.

3.3. Model validation

The numerical simulation results are furtherly validated by comparing with the experimental results coming from literatures. Fig. 4 (a) compares the outlet water temperature and PV panel temperature of the system with experiment data of Yousef et al. [50]. The photovoltaic model size is 0.35 m × 0.245 m. Its incident solar radiation fluctuates between 628 and 955 W/m^2 with time (CR = 2). Overall, the simulation results correspond well with the observed trends. The pressure drop of the pipe as function of *L/D* and outlet temperature of MPCM slurry as

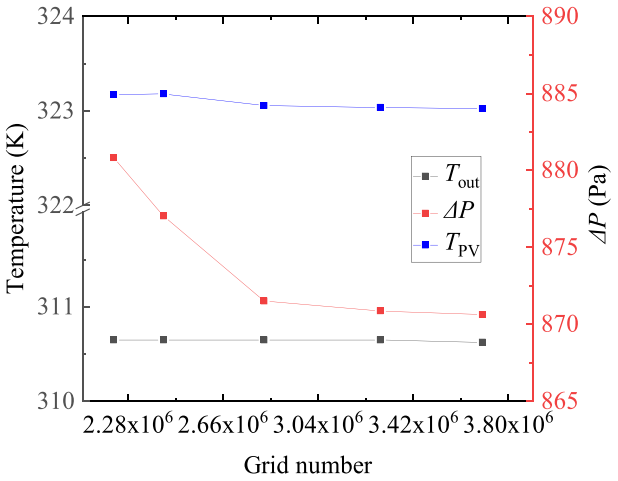
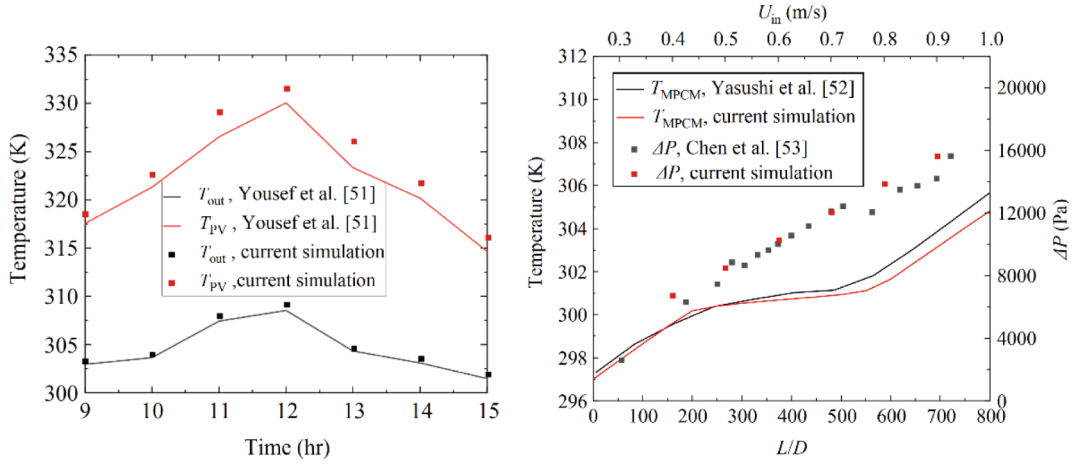


Fig. 3. Grid-independence test.

function of U_{in} are validated in Fig. 4(b) with reference to Yasushi et al. [51] and Chen et al. [52], respectively. It can be seen that the temperature and pressure drop of MPCM slurry in the tube are also in good agreement with the experimental data in literature, respectively. In the experiments conducted by Yasushi et al., MPCM slurry of $D_{pa} = 6.3$ μm and$\varphi_v = 0.12$ flowed through the experimental heating section which was a horizontal pipe with 8.0 m in length and 10.1 mm in inner diameter. Constant heat flow $Q = 5.0$ kW applied to the pipe wall. In the experiments conducted by Chen et al., MPCM slurry ($D_{pa} = 8.2$ μm, $\varphi_w =$ 15.8 wt%) flowed in the tube ($L = 1.46$ m, $D = 4.0$ mm). Fig. 4(c) compares the pressure drop and local temperature of the pipe with experiment data of Bogdan et al. [53]. Flow in a pipe, partially filled with porous material and heated from the exterior with a constant and uniform heat flux, were considered for their experimental investigation. The radius ratio and porosity of the porous media were maintained at 0.8 and 0.978, respectively. It can be seen that the numerical data are also in good agreement with the experimental data in literature.

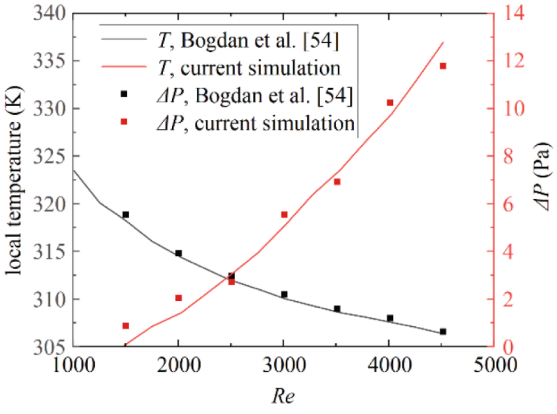
4. Results and discussion

After ensuring the accuracy and dependability of the current model, this section provides a comprehensive analysis and discussion of the effects of parameters including CPC concentration ratio, inlet flow rate, the thermophysical properties of MPCM, the thickness, and porosity of the porous medium on the thermal and electrical properties of the CPV/T system. Typical operating conditions are: $T_{in} = 310$ K, $G_{sun} = 1100$ W/m^2, $T_{amb} = 303.15$ K, and $U_{wind} = 1$ m/s.



(a) Validation of CPV/T

(b) Validation of MPCM slurry



(c) Validation of porous media

Fig. 4. Comparison with the experimental results in the literature.

4.1. The performance of the CPV/T system cooled by MPCM slurry and without porous layer

4.1.1. Effect of concentration ratio and flow rate on the CPV/T system

The effect of the CPC concentration ratio and inlet velocity of MPCM slurry on the overall performance of the CPV/T system is firstly examined in this section. Considering the actual space occupied by the concentrator and the solar radiation that the CPC can actually take, the concentration ratio examined in the study is in the range of 2.4 to 12. The inlet velocity of the MPCM slurry is 0.1 m/s − 2 m/s, which covers a range from laminar to turbulent flow. Fig. 5 shows the thermal and electrical efficiencies of the system at different concentration ratios and fluid velocities. It can be seen that the thermal efficiency of the CPV/T system increases greatly with the rise of concentration ratio as expected, while, the electrical efficiency decreases inversely due to the electrical conversion of PV module is negatively correlated with the temperature. The thermal efficiency will increase from 60.65 % to 66.03 % with the rise of concentration ratio from 2.4 to 12 when the inlet velocity of the MPCM slurry is 1 m/s, and the electrical efficiency will decrease from 10.30 % to 7.33 %, simultaneously.

It is shown in Fig. 5 that there is an optimal flow rate for both the thermal and electrical efficiency of the system at a fixed concentration ratio (CR). First, the increased flow rate allows the fluid to evacuate more heat in the same amount of time, thereby increasing the thermal and electrical efficiencies of the system. However, the increase in flow rate inevitably results in a decrease in the overall temperature of the fluid inside the tube, delaying the onset of the phase transition of the MPCM particles and, in some instances, causing the MPCM fluid to flow out of the tube without completing the phase transition process. Consequently, at this time, i.e., at a high flow rate, the flow rate continues to increase as opposed to being conducive to the enhancement of thermal efficiency. When the negative effect of the increased pressure drop in the conduit exceeds the positive effect of the decreased temperature of the PV panel, the electrical efficiency of the system decreases. The CR of the CPC has been fixed to 7.2 in all subsequent discussions as the highest overall efficiency of the system is achieved when CR = 7.2 (η_o = 72.23 %).

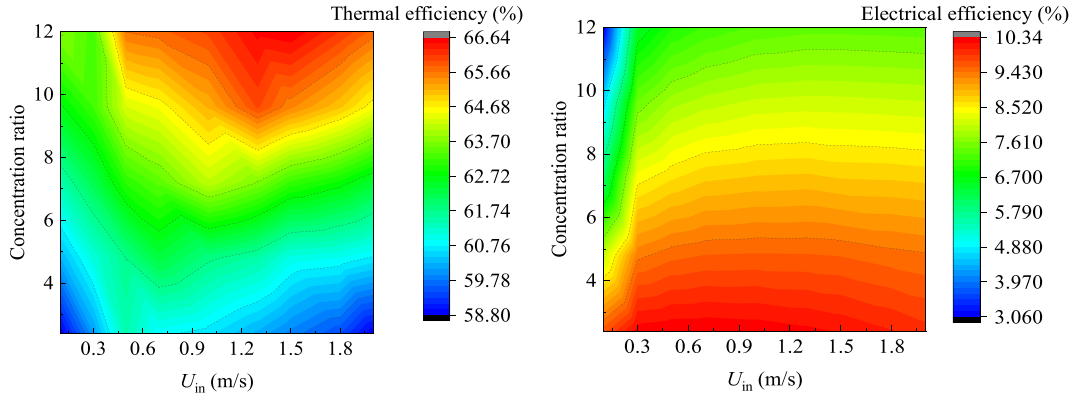
4.1.2. Effect of volume fraction of MPCM

Charunyakorn et al. [54] indicated in their study that the fluid will exhibit the characteristics of a non-Newtonian performance with unknown rheological behaviors when the MPCM volume fraction is greater than 0.2. Therefore, the volume fraction considered in this study occurs between 0.05 and 0.2. Fig. 6 and Fig. 7 show the thermal and electrical efficiencies of the system at both laminar and turbulent states at various volume fractions of MPCM slurry, respectively. Depending on the MPCM slurry volume fraction (0.05—0.2), the thermal and electrical efficiencies can be increased by 0.64 % − 7.15 % and 0.80 % − 5.45 %, respectively, when compared to the condition that pure water used as the coolant.

As the volume fraction of MPCM slurry increases, its overall specific heat capacity increases, thus enhancing the heat transfer capability of the system. However, observing Fig. 7(b), it can be clearly seen that the electrical efficiency decreases from 8.83 % to 8.79 %, and the PV panel temperature rises by 0.36 K with the rise of volume fraction of MPCM from 0.15 to 0.20. This may be because, on the one hand, according to Eq. (9), the viscosity of MPCM slurry has a strong relationship with its volume fraction. When the volume fraction in a tube is excessively high, the pressure drops increase dramatically. In addition, observing the fluid outlet temperature in Fig. 6(b), it reveals that in the turbulent state, none of the slurry with a volume fraction between 0.10 and 0.20 can endure a complete phase transition in the pipe. And with the volume fraction increases, the specific heat of solidified MPCM slurry decreases (since the specific heat of solidified MPCM particles is less than that of water). Consequently, the slurry does not absorb the heat from the PV module very well at this time. In this investigation, the optimal concentration of MPCM slurry was determined to be 0.15.

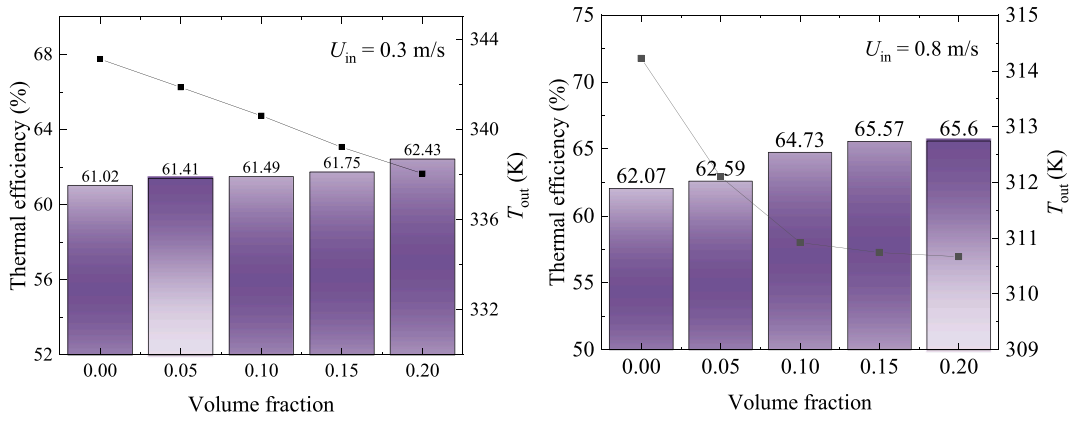
4.1.3. Effect of latent heat and melting temperature of MPCM

Fig. 8(a)–(f) illustrate the performance of the system at various latent heat and melting temperatures. It can be seen in Fig. 8(a) and (b) that the fluid outlet temperature and the PV panel temperature exhibit a decreasing and then an increasing trend with the rise of the melting temperature of MPCM, regardless of laminar or turbulent flow when the latent heat is fixed. The PV module is most sensitive to the cooling effect of the slurry when the melting temperature is 310.7 K. Considering the inlet temperature of the fluid in simulation is 310 K, it means that the suspensions with melting temperature less than 310 K are already in the process of phase change or melting at the inlet, and the latent heat of the PCM cannot be fully utilized for PV cooling. Therefore, the thermal and electrical efficiencies are relatively modest when the phase change temperature of the MPCM slurry below 310 K. Accordingly, a higher melting temperature of MPCM slurry will cause the entire system to endure phase change at a relatively high temperature, and result in an increase in fluid outlet temperature and PV panel temperature. This also causes the thermal and electrical efficiencies of the system to decrease. The thermal and electrical efficiencies of the system increase with the rise of latent heat in the laminar flow condition, where the maximal thermal and electrical efficiencies are 62.57 % and 6.17 %, respectively, as depicted in Fig. 8(c) and (e). However, when the flow rate is high (turbulent), there is a tendency for the thermal efficiency to decrease because the latent heat of slurry with the higher melting temperature



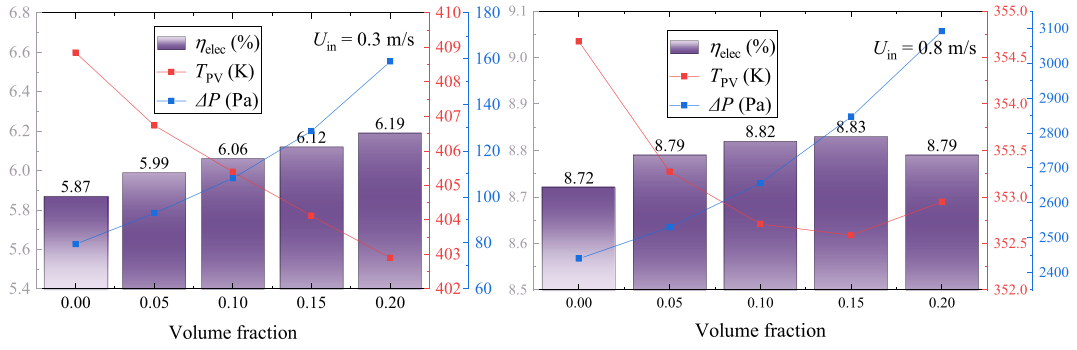
(a) Variation of thermal efficiency (b) Variation of electrical efficiency

Fig. 5. The system performance varies with concentration ratio and inlet flow rate.



(a) Laminar flow condition (b) Turbulent flow condition

Fig. 6. The system performance varies with volume fraction of MPCM slurry.



(a) Laminar flow condition (b) Turbulent flow condition

Fig. 7. The system performance varies with volume fraction.

can't be used completely for cooling as it is still under phase change process as it flows out of the tube. The selection MPCM slurry with reasonable latent heat and melting point should be accorded to the actual operating conditions of the system, and ensure that MPCM particles can increase the latent heat of the particles, assuming a complete phase change during the cooling process.

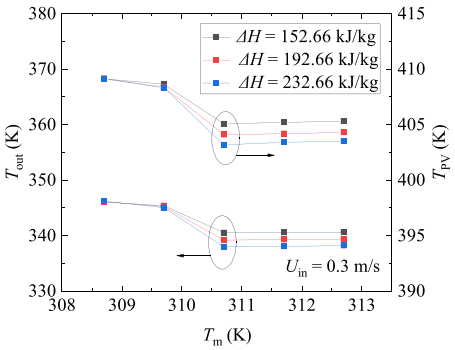
4.2. The performance of MPCM slurry based CPV/T system with partially filled porous media

4.2.1. Effect of porous layer thickness under different filling methods

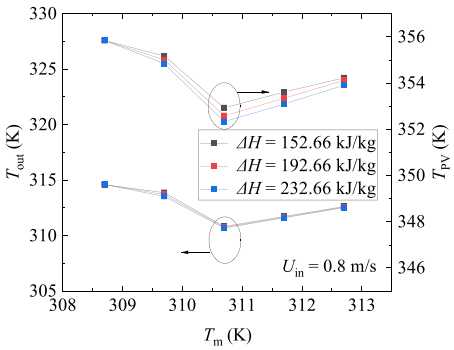
Two types of partially filling of copper foam in cooling tube are examined in this section, along the tube wall (M0 to M4 in Fig. 9) and along the tube axis (M6 to M9 in Fig. 9). The pore size and porosity of the porous media are 0.001 m and 0.5, respectively, while the MPCM slurry flows through the pipe at 0.3 m/s (laminar flow state). It is shown that both the thermal and electrical efficiencies of the system increase with the rise of the thickness of the copper foam when it's not larger than the half interior diameter of the tube along the tube wall. The presence of copper foam can enhance the heat transfer between the tube and the fluid, elevate the fluid outlet temperature, and lower the PV plate temperature. As the thickness of copper foam increases, the porous media can absorb the heat from the PV module more effectively due to its direct contact with the pipe, resulting in a continuous decrease in TPV. But since flow resistance also rises in proportion to porous media thickness, the electrical efficiency of system drops dramatically when the porous media covers the entire pipe (M5), falling by as much as 11.57 % compared to the efficiency of smooth pipe. However, it is worth noting that a steady rise in the thickness of porous medium is not beneficial to the thermal efficiency of system. Thermal efficiencies of M4 and M5 decreased by 0.20 % and 20.94 %, respectively, in comparison to M3. This trend is consistent with the research results of Milad et al. [55] and Nimvari et al. [56].

With increase of the thickness of copper foam along the tube axis (M6 to M9), the electrical efficiency of the system follows a similar pattern to that of along the tube wall. It exhibits an upward trend firstly and downward later for the similar reason mentioned above. Particularly for M9, the pressure drop inside the pipe increases apparently to almost the same level as in the case where the porous medium spreads throughout the pipe (M5) because its porous medium has almost completely spread throughout the pipe, and results in a significant decrease of the electrical efficiency compared to other cases. However, the thermal efficiency of the system is apparently different when the copper foam filled along the tube axis. It totally decreases with rise the thickness of copper foam. It can be seen in Fig. 9 that the outlet temperature and thermal efficiency decrease by 0.58 K – 2.83 K and 3.32 % – 18.32 %, respectively from M6 to M9.

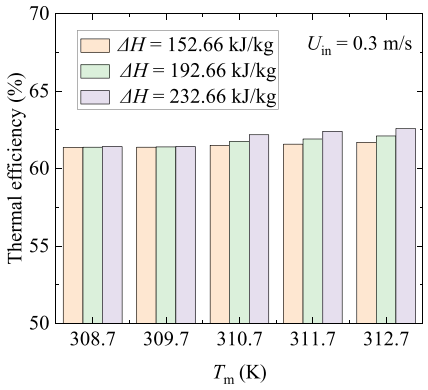
Fig. 10 illustrates the velocity and temperature distributions of the fluid along the radial direction for various models (at outlet of the pipe). It is clearly demonstrated that more fluid will be concentrated in the non-porous zone as it flows down the tube due to the reduced resistance in this area. Consequently, the fluid velocity in the porous zone is substantially lower than that in the non-porous zone. With the increase of the thickness of copper foam, the flow velocity in the non-porous zone



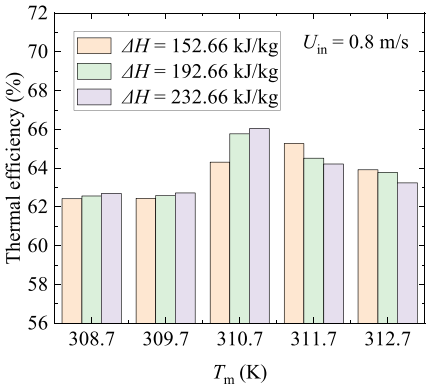
(a) T_{out} and T_{PV} in laminar flow state



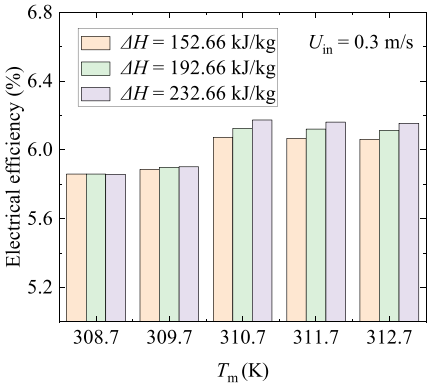
(b) T_{out} and T_{PV} in turbulent flow state



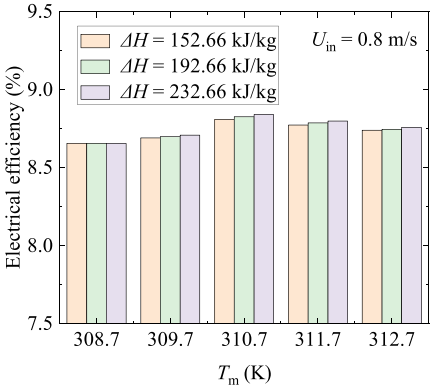
(c)Thermal efficiency in laminar flow state



(d) Thermal efficiency in turbulent flow state



(e) Electrical efficiency in laminar flow state

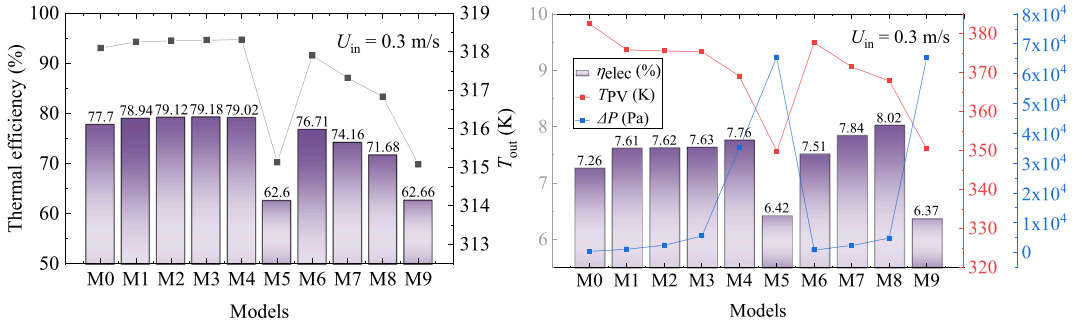


(f) Electrical efficiency in turbulent flow state

Fig. 8. The system performance varies with MPCM melting point and latent heat.

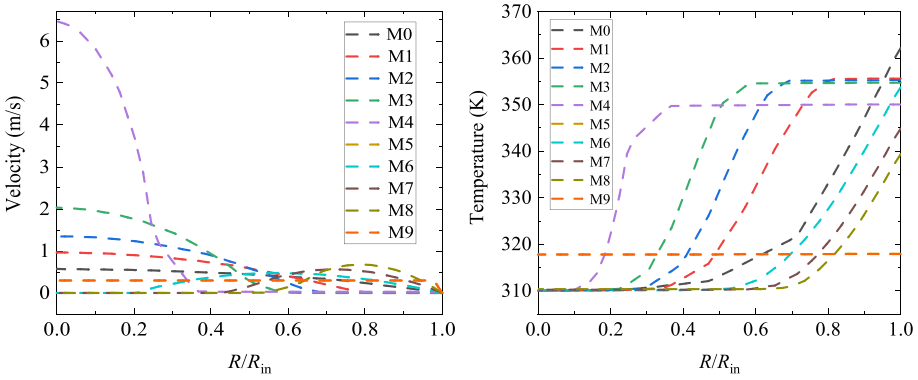
increases greatly and decreases in porous zone. This is particularly apparent for the M4. Due to the higher resistance in the porous zone, the fluid velocity is maintained at a lower level in all of the cases. Fig. 10(b) reveals that the fluid is efficiently heated when the copper foam filled along the tube wall instead of tube axis and the smooth pipe. Typically, the temperature distribution of M9 has been essentially consistent with that of a porous medium extending throughout the entire pipe of M5.

It can be seen that it has a greater ability to absorb heat from the PV module when the copper foam filled along the tube wall, and results in a significantly higher outlet water temperature and thermal efficiency compared with the same thickness of copper foam along the tube axis. The thicknesses of copper foam in M3 and M7 is identical (2.38 mm), but the outlet temperature and thermal efficiency of the former are 0.98 K and 6.34 % higher than those of the latter, respectively. Regarding the electrical performance of the system and associated parameters, the situation is distinct. As shown in Fig. 9(b), with the exception of M9 (a porous medium that almost completely covers the tube), the pressure drop is always slightly lower when the copper foam filled along the tube axis compared with that along the tube wall when the thickness of copper foam is identical. With increase of the thickness of copper foam, the cooling effect of PV panels by filling along tube axis is apparently effective than that along tube wall. Therefore, if the cooling performance of the PV module is the only consideration, the model with copper foam filled along the tube axis with the appropriate thickness



(a) Thermal efficiency and T_{out} (b) Electrical efficiency, T_{PV} and ΔP

Fig. 9. The system performance varies with porous media layer thickness for different filling methods.



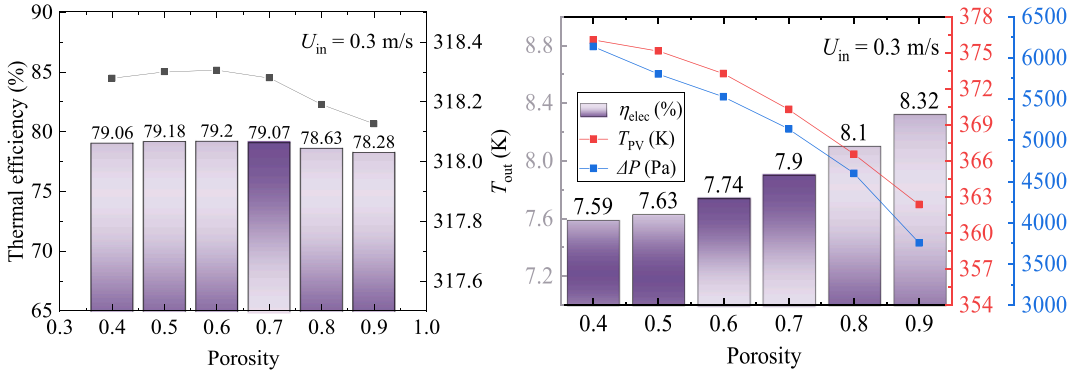
(a) Velocity distributions (b) Temperature distributions

Fig. 10. Velocity and temperature distributions in the tube along the radial direction for different models.

should be adopted. After careful consideration, the M3 model with the highest overall efficiency (86.81 %) is selected for further study because the overall performance of the CPV/T system is main concern of this research.

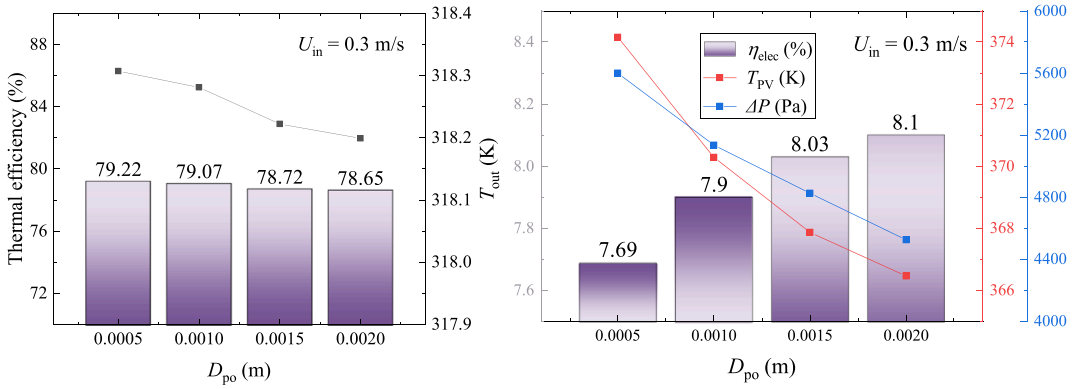
4.2.2. Effect of the porosity and pore size of copper foam

Porosity and pore size are important parameters of porous media. Figs. 11 and 12 show the effect of porosity and pore size on the performances of the system, respectively. It can be seen that the fluid outlet temperature and thermal efficiency increase firstly, then decrease with the rise of porosity, and reach the maximum values of 318.31 K and 79.20 %, respectively, when ε = 0.6. According to Eq. (4), The increase of porosity results in an increase of the effective thermal conductivity. In addition, the increase of porosity also results in an increase in the flow velocity in the porous zone, and enhances the convective heat transfer. However, when the porosity increases from 0.6 to 0.9, the outlet temperature and thermal efficiency of the fluid decrease. This may be because as the porosity increases, the temperature in the non-porous zone decreases and the temperature in the porous zone increases. Since the fluid is still mainly concentrated in the non-porous zone at this



(a) Thermal efficiency and T_{out} (b) Electrical efficiency, T_{PV} and ΔP

Fig. 11. The effect of porosity of copper foam on the system performance.



(a) Thermal efficiency and T_{out} (b) Electrical efficiency, T_{PV} and ΔP

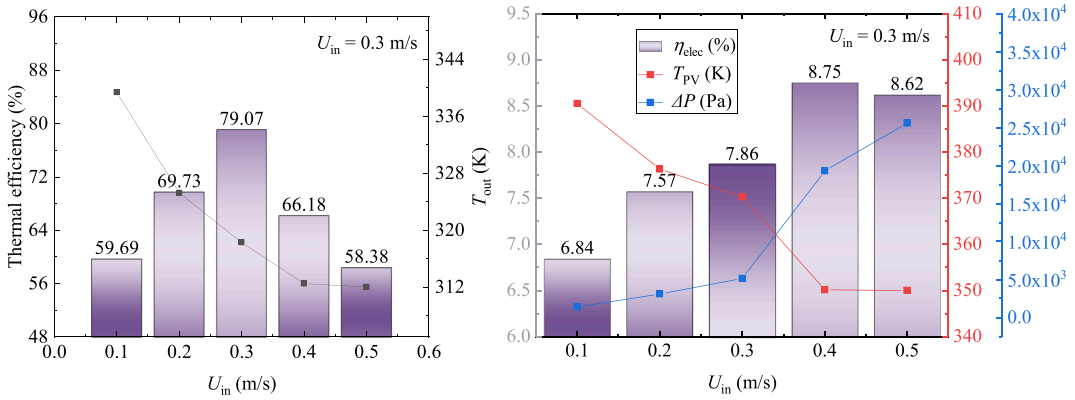
Fig. 12. The effect of pore size of copper foam on the performance of CPV/T system.

time, it leads to a decrease in the overall outlet temperature of the fluid. Fig. 11(b) indicates that an increase in porosity is always beneficial for the system electrical performance. With the increase of porosity, the PV panel temperature and pressure drop of pipes decrease significantly, that's very helpful for the enhancement of the system electrical performance. In combined consideration of the thermal and electrical performance of the system, the model with the highest overall efficiency, ε = 0.7 (87.07 %), is suggested and adopted for further analysis.

The increase of pore diameter implies an increase in permeability of porous medium. The viscous and inertial resistances also decrease with the rise of the pore diameter as demonstrated by Eqs. (10) and (11). This is why the pressure reduces with the pore size increases as shown in Fig. 12(b). Fig. 12 also illustrates the effect of pore size on the thermal and electrical efficiency of the CPV/T system. With the pore size increases, the flow rate in the non-porous region decreases, and the flow rate in the porous region increases. As the results, the fluid outlet temperature and thermal efficiency of the system tend to diminish, and the electrical efficiency continues to rise due to the decrease in PV plate temperature and pressure drop. With the pore size increases from 0.0005 m to 0.002 m, the thermal efficiency of the system declines by 0.72 % while the electrical efficiency rises by 5.33 %. The CPV/T model with D_{po} = 0.001 m is chosen for further analysis, and the overall efficiency of the system can reach its maximum value of 87.07 %.

4.2.3. *Effect of the inlet velocity of MPCM slurry*

Fig. 13 depicts the effects of the inlet flow rate of the phase change microcapsule suspension on the system performance under the porous media lay-up method of M3. When the inlet speed increased from 0.1 m/s to 0.3 m/s (laminar flow state), the thermal efficiency increases from 59.69 % to 79.17 %, and the electrical efficiency increases from 6.84 % to 7.86 %. However, the thermal efficiency decreases significantly when the flow state of MPCM slurry changes from laminar to turbulent flow. The thermal efficiency decreases by 26.26 % when the fluid inlet velocity increases from 0.3 m/s to 0.5 m/s. This may be because that the increase in fluid flow rate delays the phase change of MPCM particles, and the MPCM slurry does not complete the phase change at the outlet of the tubes when U_{in} = 0.5 m/s. It can be seen in Fig. 13(b) that the electrical efficiency is still increasing due to the significant decrease in PV plate temperature from 370.28 K to 350.11 K when velocity rise from 0.3 m/s to 0.4 m/s. However, with the further increase of flow rate from 0.4 m/s to 0.5 m/s, the flow resistance and pressure drop will increase significantly, and the advantage brought about by the PV panel's decreasing temperature can no longer outweigh the disadvantage brought about by the increasing pressure drop, resulting in a decrease in electrical efficiency from 8.75 % to 8.62 % eventually. Fig. 14 depicts the fluid velocity distribution at the channel outlet for various inlet flow rates. When the fluid flow state in the pipe changes from laminar to turbulent, there is a significant increase in the flow velocity in the porous zone, while the flow velocity in a portion of the non-porous zone is lower than in the case of U_{in} = 0.3 m/s. This means that more fluid is pushed to the pipe wall and can therefore cool the PV panel more efficiently.



(a) Thermal efficiency and T_{out} (b) Electrical efficiency, T_{PV} and ΔP

Fig. 13. The effect of the fluid inlet flow rate on the system performance.

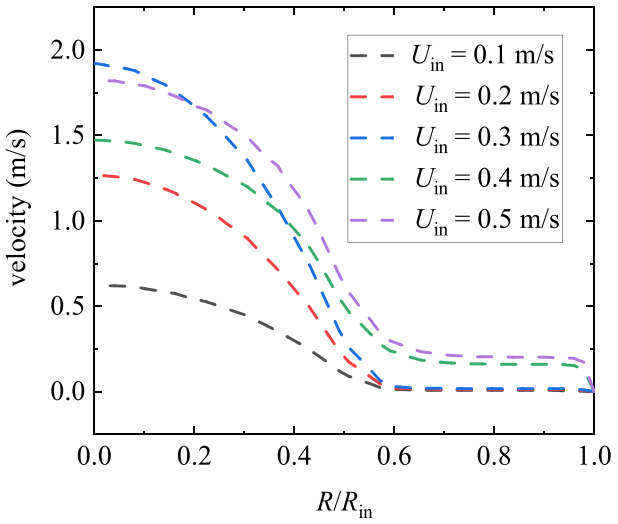


Fig. 14. Fluid velocity at the outlet of the channel at different inlet flow rates.

4.2.4. *Effect of gradient porosity and pore size*

Using the M3 model, gradient porous materials are studied to further enhance the system's performance. The porosity and pore size varied from 0.6 – 0.9 and 0.0005—0.002 m, respectively. Research is conducted on four different types of gradient porous materials, including those with porosity decreasing and pore size increasing (PDSI), decreasing porosity and decreasing pore size materials (PDSD), porosity increasing and pore size increasing (PISI), and porosity increasing and pore size decreasing (PISD). Fig. 15 compares the aforesaid four materials to a constant porosity medium ($\varepsilon = 0.75$, $D_{po} = 0.00125$ m) in terms of their performance.

The pressure loss is lowered when the porosity and pore size of the porous medium decrease linearly with pipe diameter, as compared to the situation when the porosity and pore size remain constant. Simultaneously, the reduced temperature of the PV panel is a result of the enhanced heat transfer from the PV module. When porosity and pore size rise linearly throughout the diameter of the pipe, fluid outlet temperature and thermal efficiency are maximized. The explanation for this may be that, as shown in Fig. 12, mass flow rate of fluid diminishes progressively throughout the tube's diameter, and the fluid is concentrated mostly in the tube's central region. The upward trend in Fig. 11(a) and Fig. 12(a) demonstrates the enhanced heat transmission of the fluid inside the pipe. In conclusion, PISI has the best thermal efficiency (78.97 %) and total efficiency (87.02 %), whereas PDSD has the highest electrical efficiency (8.16 %).

5. Conclusions

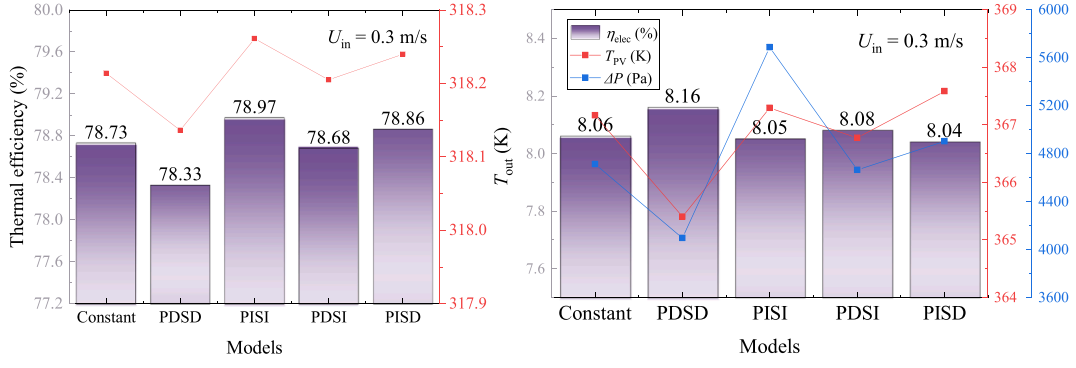
In this paper, a concentrated photovoltaic/thermal system based on composite parabolic concentrator is investigated numerically. In this system, ten pipes were uniformly laid on the back of the absorber plate. The overall efficiency of the system is improved by using micro-encapsulated phase change material slurry as a coolant in the pipes and filling the pipes with porous media in different ways. The effect of various factors on the performance of concentrated photovoltaic/thermal system is carefully examined, including the concentration ratio, the inlet velocity, and volume fraction of the fluid, the melting point and the latent heat of micro-encapsulated phase change material slurry, as well as the filling method, thickness, porosity, and pore size of the porous media. The major findings are listed below.

(1) The increase of concentration ratio of composite paraboloidal concentrator will result to the elevation of the overall system temperature, which is advantageous for thermal performance improvement, but disadvantageous for electrical efficiency promotion. There exists an optimal flow rate for the thermal and electrical efficiency of the system at a fixed concentration ratio, and this optimal flow rate progressively increases with the rise of concentration ratio.

(2) Using micro-encapsulated phase change material slurry as the heat transfer fluid of a concentrated photovoltaic/thermal system can effectively enhance the efficacy of the system. The thermal and electrical efficiencies of the system can be increased by 5.64 % and 1.26 %, respectively, when micro-encapsulated phase change material slurry with an inlet velocity of 0.8 m/s and a volume fraction of 0.15, compared to be cooled by purified water.

(3) The thermal and electrical performance of the concentrated photovoltaic/thermal system can be enhanced by increase the volume fraction at modest flow rates. However, when the fluid changes from laminar to turbulent flow, the volume fraction increases to a certain percentage due to the fluid high viscosity, which may reduce the system electrical efficiency. For the selection of micro-encapsulated phase change material particles, it should try to decrease the melting point of micro-encapsulated phase change material particles and increase the latent heat as much as possible while ensuring that the micro-encapsulated phase change material slurry can undergo a complete phase transition in the tube in order to balance the overall performance of the system.

(4) The method of partially filling the porous media along the tube wall can enhance the overall performance of the system more effectively than along the tube axis. In this manner, the greatest overall efficiency of the system is achieved to 86.81 % when the



(a) Thermal efficiency and T_{out} (b) Electrical efficiency, T_{PV} and ΔP

Fig. 15. The system performance varies with gradient porosity and pore size of porous media.

thickness of the porous media reaches half the tube diameter (4.76 mm). Increasing the porosity of the porous media facilitates the improvement of the concentrated photovoltaic/thermal system electrical efficiency. The thermal efficiency of the system decreases by 0.72 % as the pore size increases from 0.0005 to 0.002 m, while the electrical efficiency increases by 5.33 %.

(5) In the case of partially filled gradient porous media, the greatest electrical efficiency is achieved by a linear decrease in both porosity and pore diameter along the tube diameter, whereas the greatest thermal and overall efficiency is achieved by a linear increase in both porosity and pore diameter along the tube diameter.

CRediT authorship contribution statement

Gaosheng Wei: Conceptualization, Funding acquisition, Project administration, Supervision. **Yao Xiao:** Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **M. Sherin:** Investigation, Validation. **Liu Cui:** Resources, Supervision. **Xiaoze Du:** Supervision, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

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